

Deliverables

Table Relationships

learners (1) → (M) purchases

courses (1) → (M) purchases

Column	Data Type	Reason
learner_id	INT	Unique learner identifier
full_name	VARCHAR(100)	Stores learner names
country	VARCHAR(50)	Stores country names
course_id	INT	Unique course identifier
course_name	VARCHAR(100)	Stores course titles
category	VARCHAR(50)	Stores course category
unit_price	DECIMAL(10,2)	Stores monetary values
purchase_id	INT	Unique purchase identifier
quantity	INT	Number of courses purchased
purchase_date	DATE	Stores purchase date

Expected Relationships

Learner	Course
John Smith	SQL for Beginners
Priya Sharma	Python Programming
Ahmed Khan	Power BI Dashboarding
John Smith	Advanced Excel
Maria Garcia	Digital Marketing Basics
David Lee	Python Programming
Priya Sharma	Power BI Dashboarding
Ahmed Khan	SQL for Beginners

```
SELECT * FROM learners;
```

```
SELECT * FROM courses;
```

```
SELECT * FROM purchases;
```

JOINS OUTPUT

Learner_Name	Course_Name	Category	Quantity	Total_Amount	Purchase_Date
John Smith	SQL for Beginners	Database	1	199.99	2026-01-10
Priya Sharma	Python Programming	Programming	1	299.99	2026-01-15
Ahmed Khan	Power BI Dashboarding	Data Analytics	2	499.98	2026-02-01
Maria Garcia	Digital Marketing Basics	Marketing	3	449.97	2026-03-05
David Lee	Python Programming	Programming	1	299.99	2026-03-12

Learner_Name	Course_Name	Category	Quantity	Total_Amount	Purchase_Date
Ahmed Khan	Power BI Dashboarding	Data Analytics	2	499.98	2026-02-01
Maria Garcia	Digital Marketing Basics	Marketing	3	449.97	2026-03-05
Priya Sharma	Python Programming	Programming	1	299.99	2026-01-15
David Lee	Python Programming	Programming	1	299.99	2026-03-12
Priya Sharma	Power BI Dashboarding	Data Analytics	1	249.99	2026-04-08
John Smith	SQL for Beginners	Database	1	199.99	2026-01-10
Ahmed Khan	SQL for Beginners	Database	2	399.98	2026-04-20
John Smith	Advanced Excel	Productivity	1	179.99	2026-02-10

Core Analytical Queries (Q1–Q5)

Q1. Learner_Name Country Total_Spending

Ahmed Khan	UAE	899.96
Maria Garcia	Spain	449.97
Priya Sharma	India	549.98
John Smith	USA	379.98
David Lee	Singapore	299.99

Q2. Find the top 3 most purchased courses by quantity.

Course_Name	Category	Total_Quantity_Purchased
SQL for Beginners	Database	3
Power BI Dashboarding	Data Analytics	3
Digital Marketing Basics	Marketing	3

Q3. Show each category's:

- Total revenue
- Number of unique learners

Category	Total_Revenue	Unique_Learners
Data Analytics	749.97	2
Programming	599.98	2
Marketing	449.97	1
Database	599.97	2
Productivity	179.99	1

Q4. List learners who purchased from more than one category.

Learner_Name	Country	Categories_Purchased
John Smith	USA	2
Priya Sharma	India	2
Ahmed Khan	UAE	2

Q5. Identify courses never purchased.

Course_ID Course_Name Category Unit_Price

No rows returned

Q6. Find learners whose total spending is above the average learner spending

Learner_Name		Total_Spending
Ahmed Khan		899.96
Priya Sharma		549.98
Maria Garcia		449.97
John Smith		379.98
David Lee		299.99
Learner_Name	Country	Total_Spending
Ahmed Khan	UAE	899.96
Priya Sharma	India	549.98

Average Learner Spending

$899.96+549.98+449.97+379.98+299.99/5=515.98$

Q7. Display courses whose price is higher than any course in the 'Beginner' category

Course_Name	Category	Unit_Price
SQL Basics	Beginner	100.00
Excel Fundamentals	Beginner	150.00
Python Programming	Intermediate	299.99
Power BI Dashboarding	Advanced	249.99
Data Science	Advanced	399.99

Q8 . Find learners who spent more than the average spending in their country.

Learner_Name Country Total_Spending

No rows returned

Q9. Use a CTE to calculate total spending per learner, then:
Display learners with spending above 10,000.

Learner_ID Learner_Name Country Total_Spending

No rows returned

Q10. CASE Expression

Classify learners based on spending:

- Above 15,000 → "High Value",
- 8,000–15,000 → "Medium Value",
- Below 8,000 → "Low Value".

Learner_ID	Learner_Name	Country	Total_Spending	Customer_Category
3	Ahmed Khan	UAE	899.96	Low Value
2	Priya Sharma	India	549.98	Low Value
4	Maria Garcia	Spain	449.97	Low Value
1	John Smith	USA	379.98	Low Value
5	David Lee	Singapore	299.99	Low Value

Q11 . NULL Handling

- Display all courses and replace NULL purchase counts with 0 using:
`IFNULL()` or `COALESCE()`

Course_ID	Course_Name	Category	Purchase_Count
1	SQL for Beginners	Database	3
3	Power BI Dashboarding	Data Analytics	3
4	Digital Marketing Basics	Marketing	3
2	Python Programming	Programming	2
5	Advanced Excel	Productivity	1

Q12 . View

- Create a view: `category_performance_view`
- Showing:
 - *Category*
 - *Total revenue*
 - *Number of purchases*
 - *Average revenue per purchase*

Category	Total_Revenue	Number_of_Purchases	Average_Revenue_Per_Purchase
Data Analytics	749.97	2	374.99
Programming	599.98	2	299.99
Database	599.97	2	299.99
Marketing	449.97	1	449.97
Productivity	179.99	1	179.99